Department of Robotics and Automation
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# OBJECT DETECTION AND DEPTH ESTIMATION USING D435i

॥ वसुधैव कुटुम्बकम् ॥


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# INTRODUCTION

1. The ability to accurately detect objects and estimate their depth in real-time is crucial for applications such as autonomous navigation, manipulation, and environment mapping.

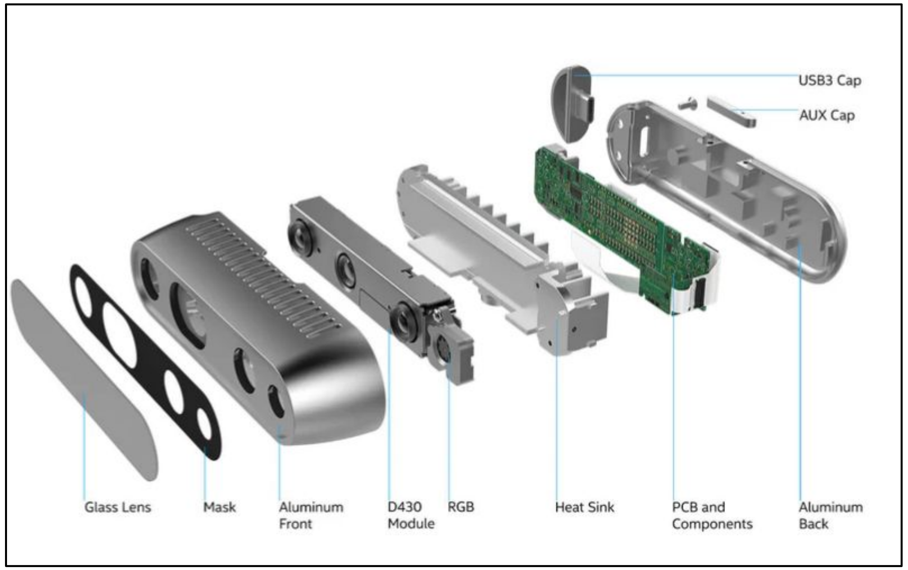
2. Traditional 2D systems lack depth awareness, leading to poor spatial perception.

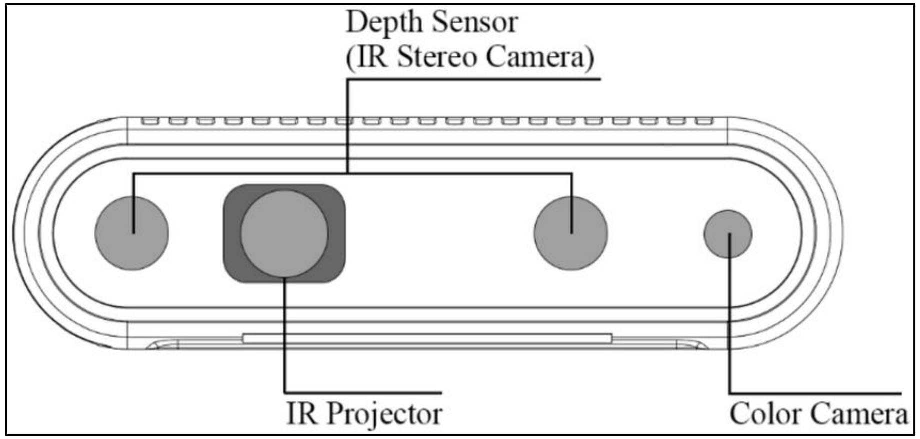
3. This project integrates a RealSense D435i camera with ROS 2, OpenCV, and YOLOv8n to detect objects and estimate their distances in real-time.

4. The Intel® RealSense™ Depth Camera D435i is an advanced depth-sensing device that integrates an Inertial Measurement Unit (IMU) to provide enhanced spatial and motion data.

# SPECIFICATIONS OF D435i

1. Depth Technology - Active Stereoscopic
2. Operating Range (Min-Max) - ~ .3m - 3m
3. Depth Resolution and FPS - 1280x720, 30fps
4. Depth Field of View - H:87 V:58
5. RGB Sensor - Yes
6. Tracking Module - N
7. Dimensions - 90mx25mx25m (Camera)
8. System Interface Type - USB 3



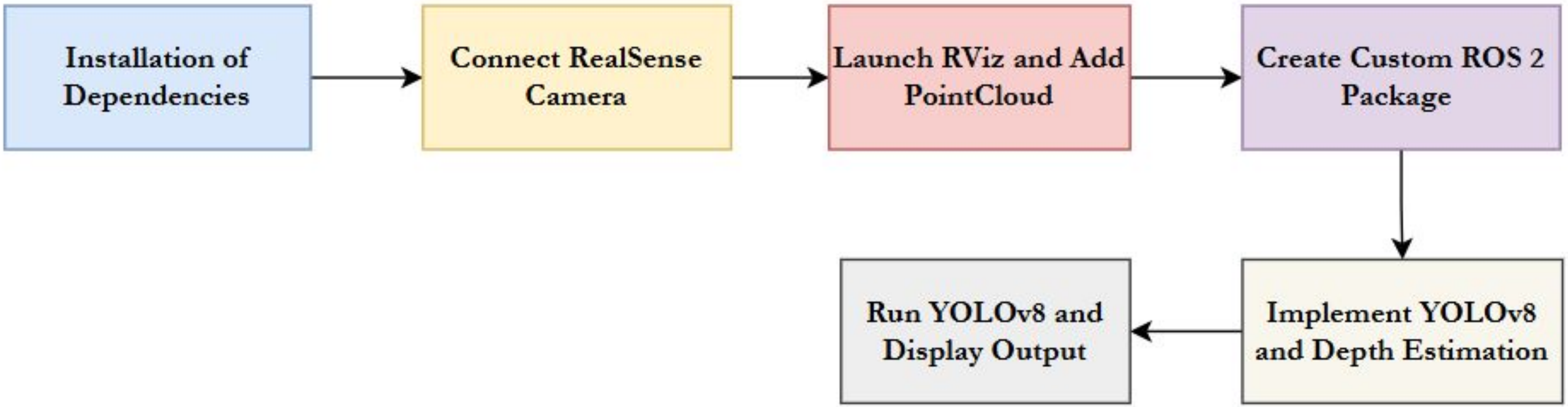




Specifications of D435i

# OBJECTIVES

1. To integrate the Intel RealSense D435i camera with ROS2 (Humble) for real-time depth and image data processing.
2. To implement object detection using OpenCV for identifying obstacles in the environment.
3. To estimate the distance of detected objects using aligned depth data from the RealSense camera.
4. To visualize the robot’s environment and navigation path using RViz.

# METHODOLOGY




Methodology followed in the study

# HARDWARE AND SOFTWARE SPECIFICATIONS

| Component | Specification / Tool |
| --- | --- |
| Camera | Intel RealSense D435i |
| Operating System | Ubuntu 22.04 |
| ROS Version | ROS 2 Humble Hawksbill |
| Programming Language | Python 3 (rclpy) |
| Libraries | OpenCV, Ultralytics YOLOv8, cv_bridge |
| Visualization | RViz2 |
| | |

# PACKAGE STRUCTURE

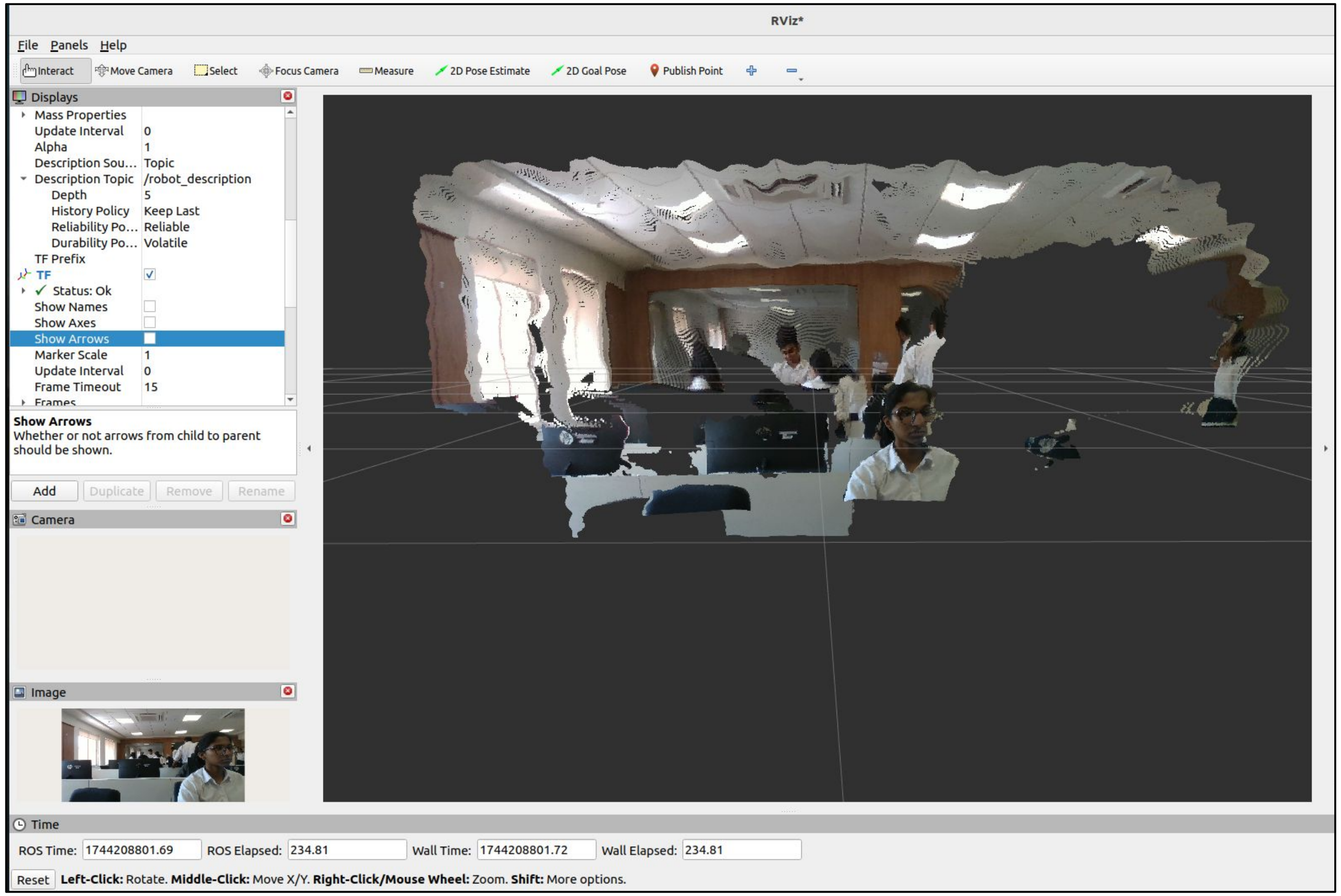
```
realsense_yolo/
├── package.xml              # Describes the package (name, dependencies, etc.)
├── setup.py                 # Python build script for installing the package
├── realsense_yolo/          # Python module directory (same name as package)
│   └── realsense_yolo_node.py # Main Python node for YOLO + depth estimation
└── launch/
    └── realsense_yolo_launch.py # Launch file to start the node via ros2 launch
```

# POINTCLOUD

- A PointCloud is a collection of data points defined in a three-dimensional coordinate system, where each point represents a specific location on the surface of an object or within an environment.

- These points are typically defined by their X, Y, and Z coordinates, capturing the spatial dimensions of the subject.

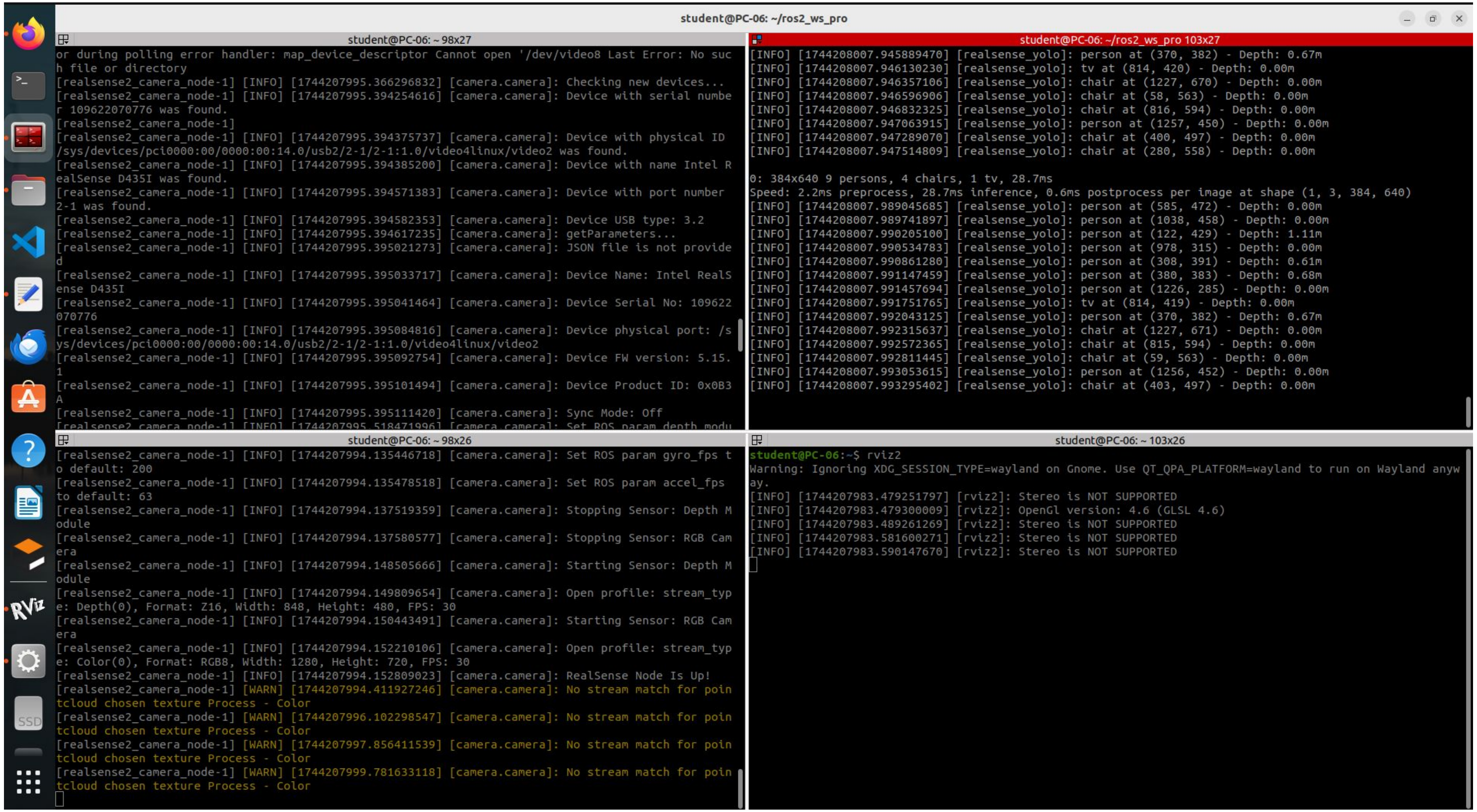
- Point clouds are commonly generated using 3D scanning technologies such as LiDAR (Light Detection and Ranging), photogrammetry, or other 3D imaging methods.



# PROCEDURE

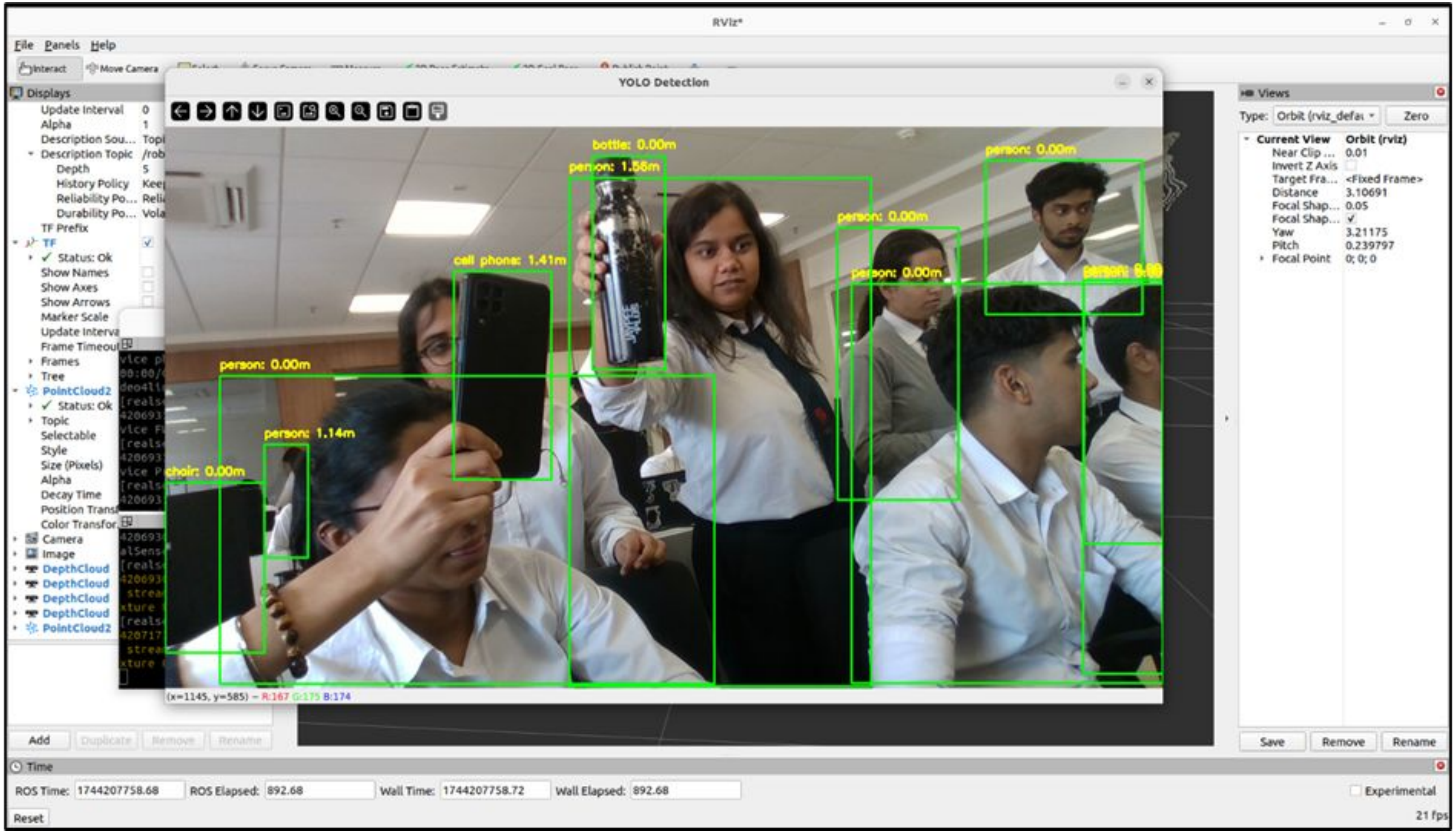
1. Source the ROS 2 environment - `source /opt/ros/humble/setup.bash`
2. Source your ROS 2 workspace - `source install/setup.bash`
3. Open RViz2 for 3D visualization - `rviz2`
4. Launch the RealSense D435i camera - `ros2 launch realsense2_camera rs_launch.py pointcloud.enable:=true`
5. Create a custom ROS 2 package - `ros2 pkg create realsense_yolo --build-type ament_python --dependencies rclpy sensor_msgs cv_bridge`
6. Build your ROS 2 workspace - `colcon build`
7. Write and run your custom YOLO + Depth node - `ros2 run realsense_yolo realsense_yolo_node`
8. Create and use a launch file for automation - `ros2 launch realsense_yolo realsense_yolo_launch.py`
9. List all active topics - `ros2 topic list`
10. View node communication graph - `rqt_graph`

# PROCEDURE

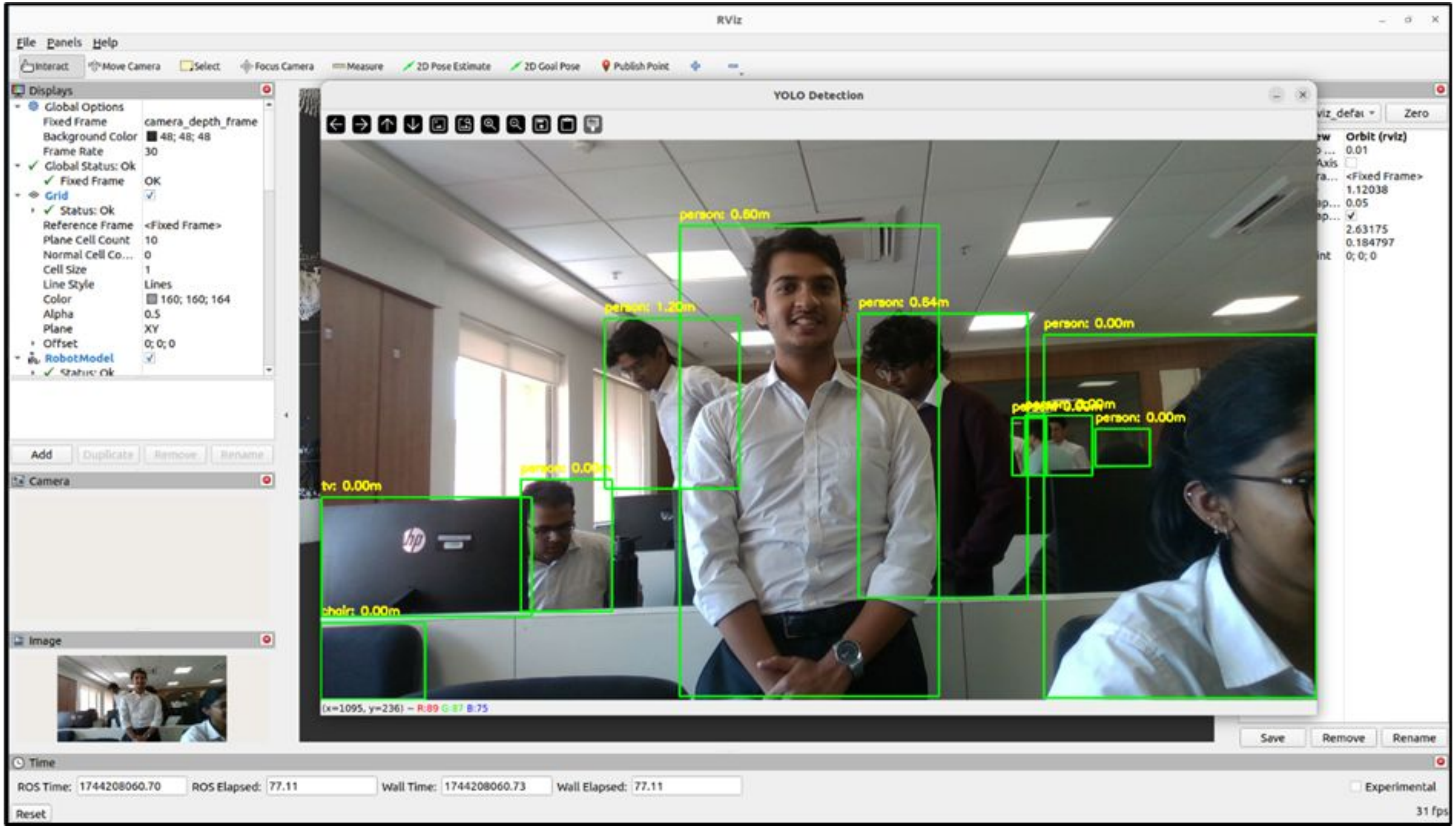


student@PC-06: ~/ros2_ws_pro
student@PC-06: ~ 98x27
or during polling error handler: map_device_descriptor Cannot open '/dev/video8 Last Error: No suc
h file or directory
[realsense2_camera_node-1] [INFO] [1744207995.366296832] [camera.camera]: Checking new devices...
[realsense2_camera_node-1] [INFO] [1744207995.394254616] [camera.camera]: Device with serial numbe
r 109622070776 was found.
[realsense2_camera_node-1]
[realsense2_camera_node-1] [INFO] [1744207995.394375737] [camera.camera]: Device with physical ID
/sys/devices/pci0000:00/0000:00:14.0/usb2/2-1/2-1:1.0/video4linux/video2 was found.
[realsense2_camera_node-1] [INFO] [1744207995.394385200] [camera.camera]: Device with name Intel R
ealSense D435I was found.
[realsense2_camera_node-1] [INFO] [1744207995.394571383] [camera.camera]: Device with port number
2-1 was found.
[realsense2_camera_node-1] [INFO] [1744207995.394582353] [camera.camera]: Device USB type: 3.2
[realsense2_camera_node-1] [INFO] [1744207995.394617235] [camera.camera]: getParameters...
[realsense2_camera_node-1] [INFO] [1744207995.395021273] [camera.camera]: JSON file is not provide
d
[realsense2_camera_node-1] [INFO] [1744207995.395033717] [camera.camera]: Device Name: Intel RealS
ense D435I
[realsense2_camera_node-1] [INFO] [1744207995.395041464] [camera.camera]: Device Serial No: 109622
070776
[realsense2_camera_node-1] [INFO] [1744207995.395084816] [camera.camera]: Device physical port: /s
ys/devices/pci0000:00/0000:00:14.0/usb2/2-1/2-1:1.0/video4linux/video2
[realsense2_camera_node-1] [INFO] [1744207995.395092754] [camera.camera]: Device FW version: 5.15.
1
[realsense2_camera_node-1] [INFO] [1744207995.395101494] [camera.camera]: Device Product ID: 0x0B3
A
[realsense2_camera_node-1] [INFO] [1744207995.395111420] [camera.camera]: Sync Mode: Off
student@PC-06: ~/ros2_ws_pro 103x27
[INFO] [1744208007.945889470] [realsense_yolo]: person at (370, 382) - Depth: 0.67m
[INFO] [1744208007.946130230] [realsense_yolo]: tv at (814, 420) - Depth: 0.00m
[INFO] [1744208007.946357106] [realsense_yolo]: chair at (1227, 670) - Depth: 0.00m
[INFO] [1744208007.946596906] [realsense_yolo]: chair at (58, 563) - Depth: 0.00m
[INFO] [1744208007.946832325] [realsense_yolo]: chair at (816, 594) - Depth: 0.00m
[INFO] [1744208007.947063915] [realsense_yolo]: person at (1257, 450) - Depth: 0.00m
[INFO] [1744208007.947289070] [realsense_yolo]: chair at (400, 497) - Depth: 0.00m
[INFO] [1744208007.947514809] [realsense_yolo]: chair at (280, 558) - Depth: 0.00m
0: 384x640 9 persons, 4 chairs, 1 tv, 28.7ms
Speed: 2.2ms preprocess, 28.7ms inference, 0.6ms postprocess per image at shape (1, 3, 384, 640)
[INFO] [1744208007.989045685] [realsense_yolo]: person at (585, 472) - Depth: 0.00m
[INFO] [1744208007.989741897] [realsense_yolo]: person at (1038, 458) - Depth: 0.00m
[INFO] [1744208007.990205100] [realsense_yolo]: person at (122, 429) - Depth: 1.11m
[INFO] [1744208007.990534783] [realsense_yolo]: person at (978, 315) - Depth: 0.00m
[INFO] [1744208007.990861280] [realsense_yolo]: person at (308, 391) - Depth: 0.61m
[INFO] [1744208007.991147459] [realsense_yolo]: person at (380, 383) - Depth: 0.68m
[INFO] [1744208007.991457694] [realsense_yolo]: person at (1226, 285) - Depth: 0.00m
[INFO] [1744208007.991751765] [realsense_yolo]: tv at (814, 419) - Depth: 0.00m
[INFO] [1744208007.992043125] [realsense_yolo]: person at (370, 382) - Depth: 0.67m
[INFO] [1744208007.992315637] [realsense_yolo]: chair at (1227, 671) - Depth: 0.00m
[INFO] [1744208007.992572365] [realsense_yolo]: chair at (815, 594) - Depth: 0.00m
[INFO] [1744208007.992811445] [realsense_yolo]: chair at (59, 563) - Depth: 0.00m
[INFO] [1744208007.993053615] [realsense_yolo]: person at (1256, 452) - Depth: 0.00m
[INFO] [1744208007.993295402] [realsense_yolo]: chair at (403, 497) - Depth: 0.00m
student@PC-06: ~ 98x26
[realsense2_camera_node-1] [INFO] [1744207994.135446718] [camera.camera]: Set ROS param gyro_fps t
o default: 200
[realsense2_camera_node-1] [INFO] [1744207994.135478518] [camera.camera]: Set ROS param accel_fps
to default: 63
[realsense2_camera_node-1] [INFO] [1744207994.137519359] [camera.camera]: Stopping Sensor: Depth M
odule
[realsense2_camera_node-1] [INFO] [1744207994.137580577] [camera.camera]: Stopping Sensor: RGB Cam
era
[realsense2_camera_node-1] [INFO] [1744207994.148505666] [camera.camera]: Starting Sensor: Depth M
odule
[realsense2_camera_node-1] [INFO] [1744207994.149809654] [camera.camera]: Open profile: stream_typ
e: Depth(0), Format: Z16, Width: 848, Height: 480, FPS: 30
[realsense2_camera_node-1] [INFO] [1744207994.150443491] [camera.camera]: Starting Sensor: RGB Cam
era
[realsense2_camera_node-1] [INFO] [1744207994.152210106] [camera.camera]: Open profile: stream_typ
e: Color(0), Format: RGB8, Width: 1280, Height: 720, FPS: 30
[realsense2_camera_node-1] [INFO] [1744207994.152809023] [camera.camera]: RealSense Node Is Up!
[realsense2_camera_node-1] [WARN] [1744207994.411927246] [camera.camera]: No stream match for poin
tcloud chosen texture Process - Color
[realsense2_camera_node-1] [WARN] [1744207996.102298547] [camera.camera]: No stream match for poin
tcloud chosen texture Process - Color
[realsense2_camera_node-1] [WARN] [1744207997.856411539] [camera.camera]: No stream match for poin
tcloud chosen texture Process - Color
[realsense2_camera_node-1] [WARN] [1744207999.781633118] [camera.camera]: No stream match for poin
tcloud chosen texture Process - Color
student@PC-06: ~ 103x26
student@PC-06:~$ rviz2
Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyw
ay.
[INFO] [1744207983.479251797] [rviz2]: Stereo is NOT SUPPORTED
[INFO] [1744207983.479300009] [rviz2]: OpenGl version: 4.6 (GLSL 4.6)
[INFO] [1744207983.489261269] [rviz2]: Stereo is NOT SUPPORTED
[INFO] [1744207983.581600271] [rviz2]: Stereo is NOT SUPPORTED
[INFO] [1744207983.590147670] [rviz2]: Stereo is NOT SUPPORTED

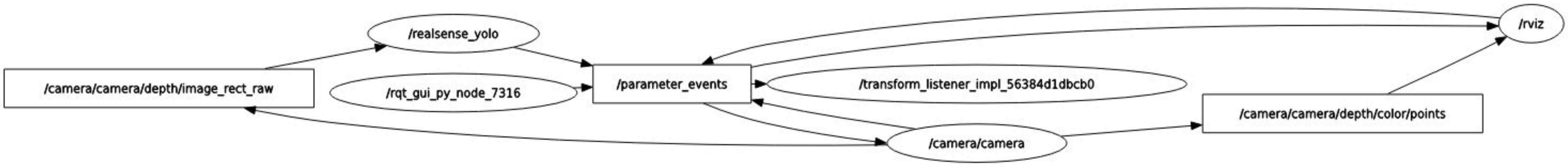
# RESULT


RViz*
File Panels Help
Interact
Move Camera
Displays
YOLO Detection
bottle: 0.00m
person: 1.56m
person: 0.00m
cell phone: 1.41m
person: 1.14m
chair: 0.00m
Views
Type: Orbit (rviz_default)
Zero
Current View
Orbit (rviz)
Near Clip ...
0.01
Invert Z Axis
Target Fra...
<Fixed Frame>
Distance
3.10691
Focal Shap...
0.05
Yaw
3.21175
Pitch
0.239797
Focal Point
0; 0; 0
Update Interval
Alpha
Description Topic
Depth
History Policy
Reliability Po...
Durability Po...
TF Prefix
TF
Status: Ok
Show Names
Show Axes
Show Arrows
Marker Scale
Frame Timeout
Frames
Tree
PointCloud2
Topic
Selectable
Style
Size (Pixels)
Decay Time
Position Transf
Color Transfor
Camera
Image
DepthCloud
Add
Duplicate
Remove
Rename
Save
Time
ROS Time: 1744207758.68
ROS Elapsed: 892.68
Wall Time: 1744207758.72
Wall Elapsed: 892.68
Experimental
Reset
21 fps

# RESULT

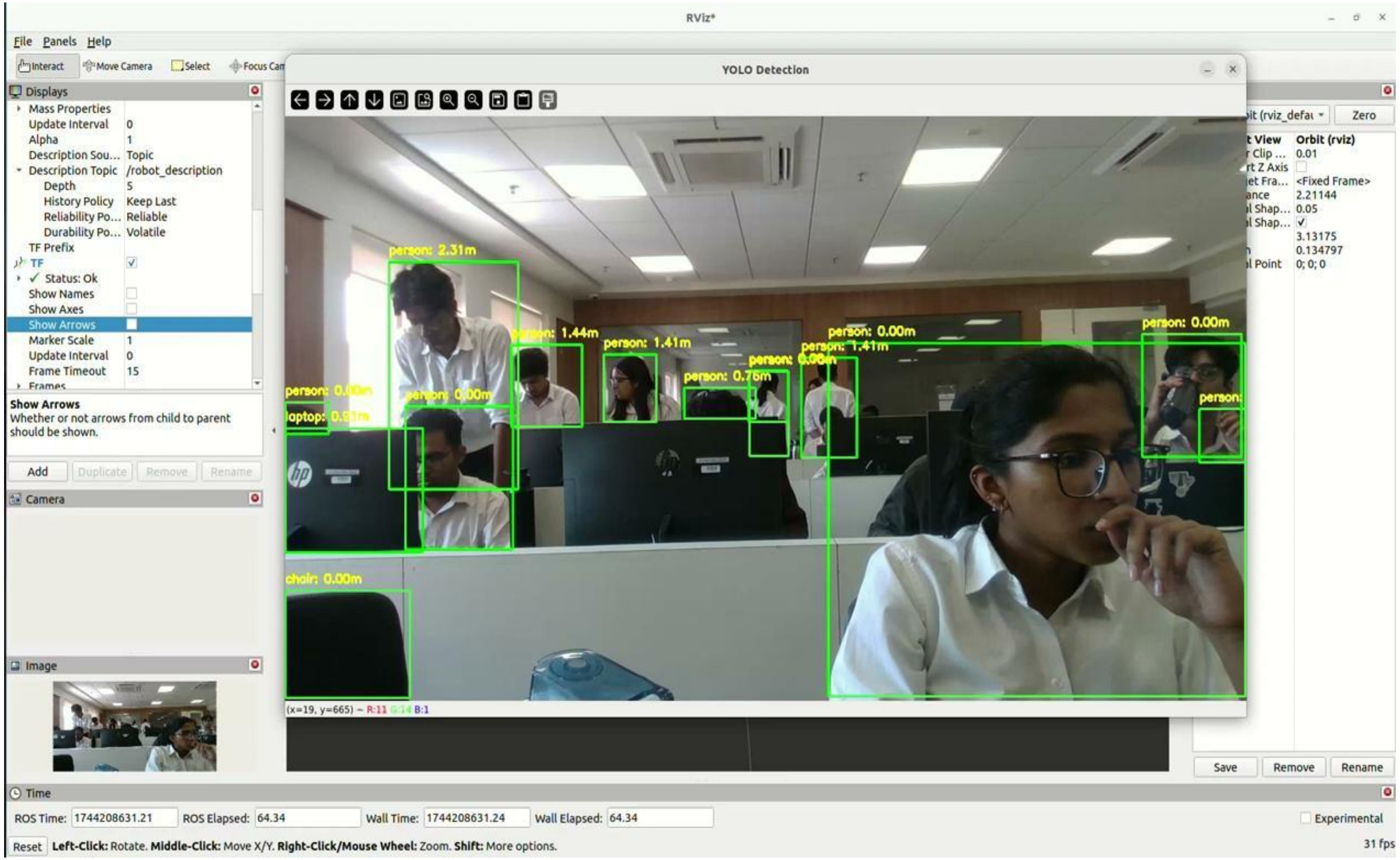

RViz
File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
Displays
Global Options
Fixed Frame camera_depth_frame
Background Color 48; 48; 48
Frame Rate 30
Global Status: Ok
Fixed Frame OK
Grid
Status: Ok
Reference Frame <Fixed Frame>
Plane Cell Count 10
Normal Cell Co... 0
Cell Size 1
Line Style Lines
Color 160; 160; 164
Alpha 0.5
Plane XY
Offset 0; 0; 0
RobotModel
Add Duplicate Remove Rename
Camera
Image
YOLO Detection
person: 0.60m
person: 1.20m
person: 0.64m
person: 0.00m
person: 0.00m
tv: 0.00m
chair: 0.00m
(x=1095, y=236) ~ R:89 G:87 B:75
Zero
Orbit (rviz)
0.01
<Fixed Frame>
1.12038
0.05
2.63175
0.184797
0; 0; 0
Save Remove Rename
Time
ROS Time: 1744208060.70
ROS Elapsed: 77.11
Wall Time: 1744208060.73
Wall Elapsed: 77.11
Experimental
Reset
31 fps

# RQT_GRAPH



/realsense_yolo
/camera/camera/depth/image_rect_raw
/rqt_gui_py_node_7316
/parameter_events
/transform_listener_impl_56384d1dbcb0
/rviz
/camera/camera/depth/color/points
/camera/camera

# WORKING VIDEO



# CONCLUSION

- The system successfully integrates object detection and depth estimation using the RealSense D435i and ROS 2.

- It accurately detects multiple objects, calculates their distances, and displays results in real time with RViz2 and OpenCV.

- The modular ROS 2 design ensures easy scalability for future robotics applications.

# THANK YOU